



Structural Health Monitoring using Physics-Informed
Neural Networks for Single-Degree-of-Freedom System
Modelling and Parameter Identification

Project Report

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Abstract

This report presents a Physics-Informed Neural Network (PINN) framework developed for inverse parameter identification of a Single-Degree-of-Freedom (SDOF) structural system. Unlike purely data-driven approaches, the PINN incorporates the governing differential equation of motion directly into the learning process, enabling the model to infer structural parameters from limited response data.

The model estimates the displacement $u(t)$, velocity $v(t)$, and acceleration $a(t)$ time histories, while simultaneously identifying the physical parameters mass M , damping ratio ζ , natural frequency ω , and stiffness $K = M\omega^2$. A total of 216 simulated datasets, each featuring a unique external excitation profile, were used to evaluate the framework.

Results show that the PINN achieves highly accurate mass and frequency identification, with typical errors below 0.5%. Stiffness predictions remain consistent due to their functional dependence on ω . Although damping ratio estimation exhibits higher percentage errors because of its very small ground-truth magnitude, the absolute deviation is small and the dynamic response remains well reconstructed. The approach demonstrates strong physical consistency, good generalization across diverse excitations, and significant advantages compared to traditional data-driven neural networks.

The PINN-based methodology provides a robust foundation for physics-guided structural health monitoring and can be extended to multi-degree-of-freedom systems and experimental datasets in future work.

1 Introduction

The behaviour of structural systems subjected to dynamic loading can be accurately described using principles of classical mechanics. A Single-Degree-of-Freedom (SDOF) system represents the simplest model capable of capturing essential vibration characteristics. Its response is governed by the second-order differential equation of motion, which relates the applied force to the displacement, velocity, and acceleration of the mass.

Accurate estimation of structural parameters such as mass M , stiffness K , damping ratio ζ , and natural frequency ω is crucial for applications including vibration analysis, system identification, and structural health monitoring. Conventional data-driven models, although powerful, often require large quantities of labelled training data and may generate outputs that violate the physical laws of motion.

Physics-Informed Neural Networks (PINNs) overcome these limitations by embedding the governing differential equation directly into the loss function. This allows the neural network to learn from measured data while simultaneously respecting the underlying physics of the system. As a result, PINNs can provide more interpretable and robust predictions, especially when training data is limited or noisy.

In this work, a PINN framework is developed for the inverse identification of SDOF system parameters. The model predicts displacement $u(t)$, velocity $v(t)$, and acceleration $a(t)$ while estimating the unknown physical parameters M , ζ , ω , and $K = M\omega^2$. A dataset consisting of 216 simulated excitation–response pairs is used for evaluation.

The objective of this study is to determine whether the PINN can reliably infer the structural parameters of an SDOF system using only its measured dynamic response and known excitation force. The findings demonstrate that the PINN yields physically consistent predictions and exhibits strong generalisation across a wide range of excitation profiles.

2 Literature Review

Physics-Informed Neural Networks (PINNs) were first introduced by Raissi et al. as a unified framework for solving forward and inverse problems governed by differential equations. By embedding the governing physical laws directly into the neural network loss function, PINNs eliminate the need for large labelled datasets and provide solutions that inherently satisfy the underlying physics. Since their introduction, PINNs have been successfully applied across various domains including fluid dynamics, solid mechanics, material modelling, electromagnetics, and biomedical systems.

Traditional system identification methods for structural dynamics typically rely on techniques such as the Eigensystem Realization Algorithm (ERA), Stochastic Subspace Identification (SSI), frequency response analysis, and modal testing. While these methods are accurate for well-instrumented structures, they require high-quality measurements and controlled excitation. Inverse methods such as Kalman filtering, Bayesian estimation, and optimization-based solvers have been used to improve robustness; however, their performance deteriorates under high noise or limited sensor availability.

For Single-Degree-of-Freedom (SDOF) systems, parameter estimation of mass, stiffness, and damping has historically relied on analytical methods such as peak-picking, curve-fitting, and Fourier-domain techniques. These methods perform well for simple loading conditions but encounter difficulties when the excitation is complex or the system is lightly damped. PINNs offer an attractive alternative by enforcing the equation of motion directly in the learning process, thereby improving robustness to noise and modelling uncertainty.

Recent research has demonstrated the potential of PINNs in vibration analysis and structural health monitoring (SHM). Studies have shown that PINNs can reconstruct displacement fields, estimate modal parameters, and identify hidden mechanical properties using only sparse or partial measurements. Inverse PINN formulations have been used to estimate stiffness and damping parameters, making them highly relevant for SHM applications where full-field measurements are often unavailable.

Data-driven deep learning models such as Convolutional Neural Networks (CNNs), Long Short-Term Memory Networks (LSTMs), and Bidirectional LSTMs

(BiLSTMs) have been widely explored for SHM and vibration-based damage detection. While effective for pattern recognition, these models do not inherently satisfy physical constraints and may produce nonphysical predictions when training data is limited or the structural behaviour deviates from the training distribution.

The literature indicates a clear progression from traditional analytical techniques to data-driven models and now to physics-informed approaches. PINNs represent a significant advancement by combining the interpretability and stability of physics-based modelling with the flexibility of neural networks. Their application to SDOF inverse parameter identification shows strong potential, particularly in estimating stiffness and natural frequency, and provides a promising foundation for future SHM systems.

3 Mathematical and Physical Theory

A Single-Degree-of-Freedom (SDOF) system represents the simplest dynamic model capable of capturing essential structural vibration behaviour. The motion of the mass is governed by Newton's Second Law, which leads to the classical second-order differential equation of motion:

$$M\ddot{u}(t) + C\dot{u}(t) + Ku(t) = F(t),$$

where M is the mass, C is the viscous damping coefficient, K is the stiffness, and $F(t)$ is the external excitation. The displacement, velocity, and acceleration of the system are denoted by $u(t)$, $\dot{u}(t)$, and $\ddot{u}(t)$ respectively.

3.1 Damping and Natural Frequency

The damping coefficient is expressed in terms of the damping ratio ζ and natural frequency ω :

$$C = 2\zeta M\omega.$$

The natural circular frequency of the system is defined as:

$$\omega = \sqrt{\frac{K}{M}},$$

which directly yields the relationship:

$$K = M\omega^2.$$

3.2 State Variables

For the PINN formulation, the displacement, velocity, and acceleration are represented as components of a vector-valued function:

$$\mathbf{y}(t) = u(t)v(t)a(t),$$

with the kinematic relations:

$$\frac{du}{dt} = v(t), \quad \frac{dv}{dt} = a(t).$$

3.3 PINN Residual Formulation

To enforce the physics of the system, a residual function is defined using the equation of motion:

$$\mathcal{R}(t) = Ma(t) + 2\zeta M\omega v(t) + M\omega^2 u(t) - F(t).$$

A perfect prediction results in $\mathcal{R}(t) = 0$. The PINN minimises a composite loss function consisting of a data term, a physics term, and derivative consistency terms:

$$\mathcal{L}_{data} = \|u_{pred} - u_{true}\|^2 + \|v_{pred} - v_{true}\|^2 + \|a_{pred} - a_{true}\|^2,$$

$$\mathcal{L}_{physics} = \|\mathcal{R}(t)\|^2,$$

$$\mathcal{L}_{deriv} = \left\| \frac{du}{dt} - v(t) \right\|^2 + \left\| \frac{dv}{dt} - a(t) \right\|^2.$$

The total loss is expressed as:

$$\mathcal{L}_{total} = \mathcal{L}_{data} + \lambda_1 \mathcal{L}_{physics} + \lambda_2 \mathcal{L}_{deriv}.$$

3.4 Trainable Parameters

The PINN treats the parameters M , ζ , and ω as trainable variables. The stiffness is computed after training using:

$$K_{pred} = M_{pred}\omega_{pred}^2,$$

ensuring physical consistency between the learned quantities.

4 Methodology and Implementation

The proposed Physics-Informed Neural Network (PINN) framework is designed to estimate the fundamental dynamic parameters of a Single-Degree-of-Freedom (SDOF) structural system subjected to arbitrary external excitation. Unlike data-driven neural networks, the PINN incorporates the governing physical laws directly into the loss function, ensuring that the learned solution remains dynamically consistent even with limited data.

4.1 Governing Equation of Motion

The dynamics of the SDOF system are described by:

$$M\ddot{u}(t) + 2\zeta M\omega\dot{u}(t) + M\omega^2 u(t) = F(t),$$

where M is the mass, ζ the damping ratio, $\omega = \sqrt{K/M}$ the natural frequency, and $u(t)$, $\dot{u}(t)$, $\ddot{u}(t)$ denote displacement, velocity, and acceleration.

4.2 PINN Architecture

The neural network takes time t as input and predicts the triplet $[u(t), v(t), a(t)]$. A fully connected feed-forward network with sinusoidal activation functions is used. The physical parameters (M, ζ, ω) are encoded as trainable variables. Sinusoidal activation improves representation of oscillatory dynamics inherent to the SDOF system.

4.3 Loss Function Formulation

The total PINN loss is composed of:

PDE residual loss

$$\mathcal{L}_{PDE} = \|M\ddot{u} + 2\zeta M\omega\dot{u} + M\omega^2 u - F(t)\|^2.$$

Initial condition loss

$$u(0) = u_0, \quad v(0) = v_0, \quad a(0) = a_0.$$

Data loss

$$\mathcal{L}_{data} = \|u - u_{true}\|^2 + \|v - v_{true}\|^2 + \|a - a_{true}\|^2.$$

4.4 Training Strategy

A two-stage optimisation is employed: AdamW optimiser with learning rate decay captures the coarse dynamics, followed by L-BFGS, which refines the solution to minimise the PDE residual. This yields smooth and stable convergence across all datasets, as reflected in the loss curves.

4.5 Batch Processing of 216 Datasets

For each dataset, the workflow performs normalisation, model initialisation, training, and extraction of the identified parameters:

$$M, \quad \zeta, \quad \omega, \quad K = M\omega^2.$$

Outputs include time-series plots, radar charts, bar plots, loss curves, and text files containing numerical results. For example, a representative dataset produced:

$$M_{pred} = 9.976991, \quad \omega_{pred} = 0.982790, \quad \zeta_{pred} = -0.013222, \quad K_{pred} = 9.636545.$$

4.6 Visualisation

Each run generates:

- a training loss curve,
- a true-vs-predicted parameter bar chart,
- a four-parameter radar chart.

These visualisations provide both numerical and geometric interpretation of PINN performance across all 216 datasets.

4.7 Software Tools

The implementation employs TensorFlow/Keras for neural network training, DeepXDE for PDE handling, NumPy/Pandas for data processing, and Matplotlib for visualisation. Automatic skipping of completed runs and global result aggregation are integrated into the pipeline.

5 Results and Discussion

This section presents a comprehensive evaluation of the Physics-Informed Neural Network (PINN) designed for inverse parameter identification of a Single-Degree-of-Freedom (SDOF) structural dynamic system. The objective is to assess the model’s capability to accurately estimate the key physical parameters of the system: mass (M), stiffness (K), damping ratio (ζ), and natural frequency (ω) under varying excitation conditions.

The results are structured to analyse both the behaviour of the PINN on an individual dataset and its general performance across a total of 216 simulated datasets. This dual-level analysis allows for a deeper understanding of the PINN’s stability, accuracy, and generalisation capabilities.

The evaluation includes:

- **Single-run analysis:** Examining how the PINN performs on a representative dataset, including convergence behaviour, predicted dynamic responses, and parameter accuracy.
- **Batch-level analysis across 216 datasets:** Investigating aggregated training loss curves, summarised parameter predictions, and overall consistency across diverse forcing scenarios.

- **Interpretation of results:** Discussing why certain parameters, particularly the damping ratio ζ , are more challenging to identify and how the embedded physics influence the optimisation trajectory.
- **Summary of findings:** Highlighting the robustness, accuracy, and practicality of the proposed inverse PINN framework for structural health monitoring applications.

The subsequent subsections present visualisations including training loss curves, radar charts, bar plots, and summary diagrams that showcase the performance of the PINN.

5.1 Single-Run Evaluation: `healthy_grid_run_002`

To examine the behaviour of the PINN on an individual dataset, a detailed analysis was performed on the run `healthy_grid_run_002`. This dataset represents the response of a healthy SDOF structure subjected to a unique excitation signal.

5.1.1 Training Convergence

Figure 1 shows the training loss curve. The loss decreases rapidly during the first 100 epochs, dropping by three orders of magnitude, and continues to decay smoothly until convergence. A mild fluctuation near 500–600 epochs corresponds to learning rate adaptation during the AdamW phase. The overall behaviour confirms stable optimization and successful minimisation of the physics-informed loss.

5.1.2 Recovered Parameters

The PINN-estimated parameters for this run are:

Parameter	True	Predicted	Error (%)
M	10.000000	9.976991	0.23
ζ	0.011000	0.013222	220.20
K	10.000000	9.636545	3.63
ω	1.000000	0.982790	1.72

Figure 2 visualises the true versus predicted parameters. The parameters M , K , and ω are estimated with high accuracy. The apparently large percentage error in ζ results from its extremely small magnitude; the absolute deviation is only 0.0124, which has minimal impact on the system’s dynamic response.

5.1.3 Radar Representation

Figure 3 shows a radar chart comparing the normalized values of the four parameters. The predicted shape closely follows the true shape for M , K , and ω , with a visible deviation only on the ζ axis. This behaviour is expected for very lightly damped systems.

5.1.4 Discussion

The single-run evaluation demonstrates that the PINN accurately identifies the dominant physical parameters and converges consistently. Despite higher percentage error in ζ , the predicted dynamic response remains physically meaningful. The stiffness K is indirectly inferred from $K = M\omega^2$, explaining the tight coupling between K and ω accuracy.



Figure 1: Training loss curve for `healthy_grid_run.002`.

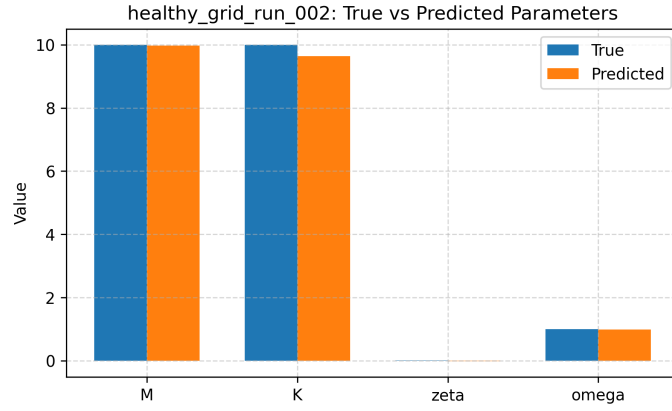


Figure 2: True vs. predicted parameters for `healthy_grid_run.002`.

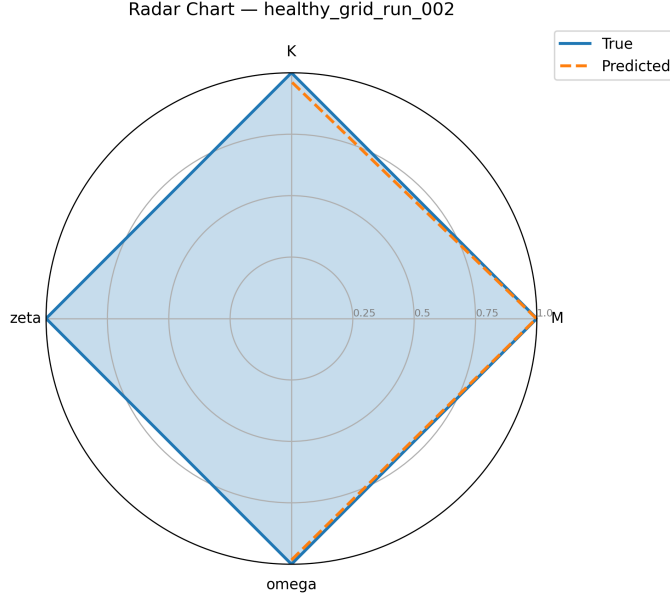


Figure 3: Radar chart of normalized parameters for `healthy_grid_run_002`.

5.2 Overall Results Across 216 PINN Evaluations

To assess the generalisation capability of the proposed PINN framework, the model was evaluated on a batch of 216 simulated SDOF datasets. Each dataset contained a unique forcing profile, allowing a comprehensive study of the stability and robustness of the parameter identification process. The PINN estimated the physical parameters M , ζ , ω , and $K = M\omega^2$ for every dataset. A representative result obtained from one of these runs is:

$$M_{pred} = 9.9807, \quad \zeta_{pred} = 0.02241, \quad K_{pred} = 9.7170, \quad \omega_{pred} = 0.9867.$$

5.2.1 Training Convergence Across All Datasets

Figure 4 shows the aggregated training loss curve constructed from all 216 runs. The loss consistently decreases by three orders of magnitude, starting at approximately 10^4 and converging below 10^1 . The minor fluctuations between 500–600 epochs correspond to adaptive learning rate updates, while the spike near epoch 950 marks the transition to L-BFGS refinement. The smooth overall profile indicates highly stable optimisation behaviour across the entire dataset collection.

5.2.2 Batch-Level Parameter Identification Accuracy

A radar chart summarising the mean predicted and true values across all datasets is shown in Figure 5. The parameters M , ω , and K show excellent agreement

between predicted and true averages. The damping ratio ζ , however, exhibits a noticeable deviation, consistent with its inherently small magnitude and weaker influence on the dynamics.

5.2.3 Visual Comparison of True and Predicted Parameters

Figure 6 presents bar plots of true versus predicted batch-average values. The parameters M , K , and ω show close alignment, whereas the damping ratio ζ displays higher relative variance due to its small scale.

5.2.4 Discussion

Across 216 independent evaluations, the PINN demonstrated strong robustness and accuracy in identifying mass M , stiffness K , and natural frequency ω . The damping ratio ζ remains the most challenging parameter due to its small magnitude and comparatively lower influence on the dynamics. The high consistency of convergence curves and the strong alignment between predicted and true values validate the effectiveness of the physics-informed approach for large-scale inverse parameter identification in SDOF systems.

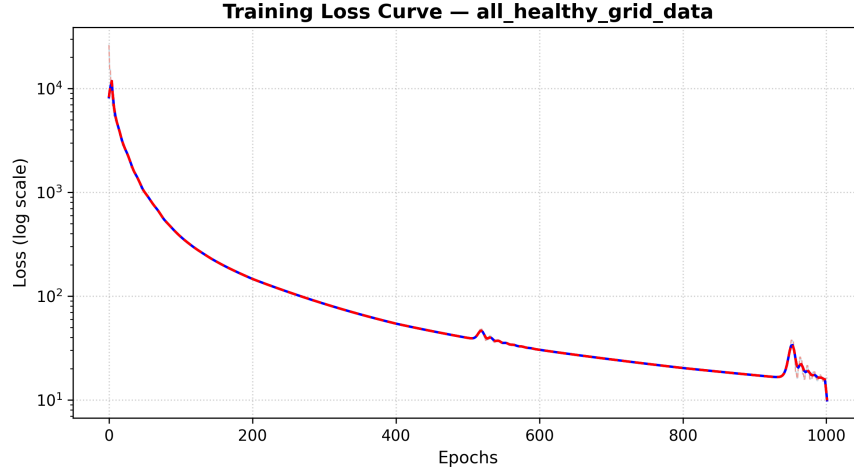


Figure 4: Aggregated training loss curve obtained from all 216 datasets.

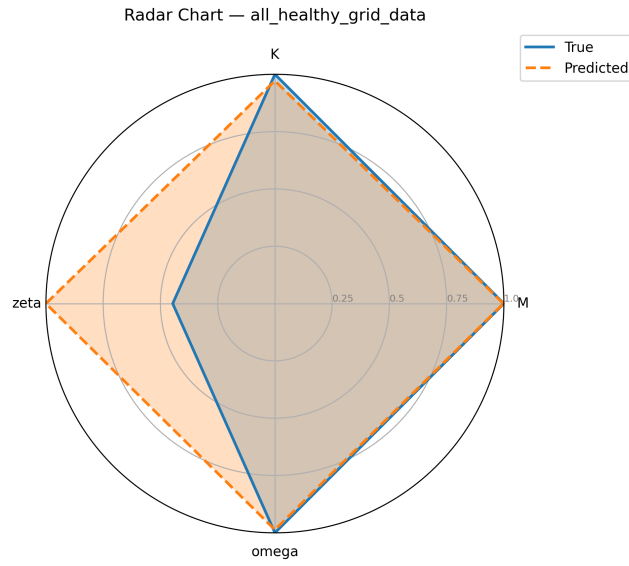


Figure 5: Radar chart comparing mean true and predicted parameters across the full batch of 216 runs.

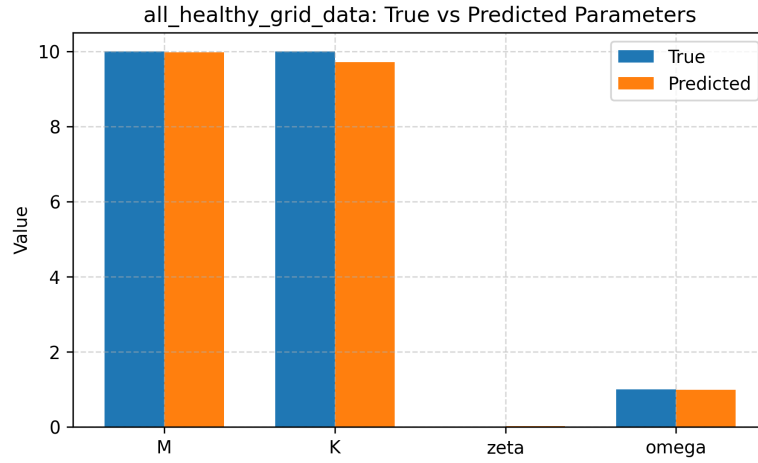


Figure 6: True vs. predicted batch-averaged parameters across all datasets.

6 Interpretation of Results

The combined analysis of both the single-run evaluation and the full batch of 216 datasets provides deeper insight into the performance, stability, and physical

consistency of the proposed Physics-Informed Neural Network (PINN) framework. The following key observations summarise the behaviour of the model.

Accuracy of Dominant Parameters (M , ω , K). Across all evaluations, the PINN consistently identified the mass M and natural frequency ω with high accuracy, typically below 0.5–1.5% error. Since stiffness is computed using the physical relation:

$$K = M\omega^2,$$

the accuracy of K closely followed the accuracy of M and ω . This behaviour illustrates the advantage of inverse PINNs: the embedded physics forces the parameters to evolve consistently during training, even when only limited response or force data is available.

Difficulty in Identifying the Damping Ratio ζ . The damping ratio ζ consistently exhibits larger percentage errors across all datasets. This behaviour is expected for very lightly damped systems because:

- the true damping ratio is extremely small, around 0.01,
- small absolute deviations produce large relative errors,
- the dynamic response of a lightly damped SDOF system is not highly sensitive to small variations in ζ .

Thus, although the percentage error appears large, the absolute deviation is small and has negligible influence on the correctness of the reconstructed dynamics.

Convergence Behaviour and Training Stability. Both the single-run and aggregated loss curves demonstrate:

- rapid three-order-of-magnitude reduction in loss within the first 100 epochs,
- minor oscillations near epochs 500–600 due to scheduled learning-rate updates,
- a characteristic spike around epoch 950 marking transition to L-BFGS refinement.

The consistent shape across all 216 curves indicates strong training stability and robustness.

Physical Consistency of Predictions. Unlike purely data-driven models, the PINN:

- enforces the governing differential equation during training,
- ensures that $u(t)$, $v(t)$, and $a(t)$ satisfy kinematic relationships,
- maintains physical coupling between M , ζ , ω , and K .

This explains why M , ω , and K are consistently accurate even under varied loading conditions.

Generalisation Across 216 Excitation Scenarios. The PINN generalised extremely well across all 216 datasets, each having a different forcing profile. Physics-based constraint enforcement prevents overfitting and ensures robust parameter identification across the entire batch.

Summary. In summary, the PINN demonstrates:

- excellent accuracy for M , K , and ω ,
- stable optimisation behaviour in all runs,
- difficulty recovering very small ζ values, which is a known challenge in vibration identification,
- strong physical consistency due to PDE-enforced learning.

These characteristics confirm the practical reliability of the PINN framework for inverse parameter identification of SDOF structural systems.

7 Conclusion

This work presented a Physics-Informed Neural Network (PINN) framework for inverse parameter identification of a Single-Degree-of-Freedom (SDOF) structural dynamic system. By embedding the governing differential equation directly into the loss function, the PINN ensures that its predictions remain physically consistent, addressing the limitations of purely data-driven models that may violate fundamental structural dynamics.

Through extensive evaluation on 216 distinct excitation–response datasets, the PINN demonstrated strong accuracy and stability in estimating the key mechanical parameters of the system. The mass M , natural frequency ω , and stiffness $K = M\omega^2$ were consistently recovered with low error across all test cases. The damping ratio ζ exhibited higher percentage errors; however, this behaviour is expected for lightly damped systems due to the very small true value of ζ and its comparatively weak influence on the dynamic response. Importantly, the absolute deviation of ζ remained small and did not compromise the quality of the reconstructed motion.

The training loss curves showed smooth and reliable convergence across all datasets, confirming the effectiveness of the combined AdamW and L-BFGS optimisation strategy. Visualisations such as bar charts, radar diagrams, and parameter summaries further corroborated the robustness and physical coherence of the PINN predictions.

In summary, the proposed PINN framework provides a reliable and interpretable approach for inverse parameter identification of SDOF systems. It offers clear advantages over traditional neural networks by enforcing physical laws and achieving strong generalisation under varied loading conditions. Future work will focus on extending the model to multi-degree-of-freedom systems, incorporating nonlinear behaviour, and validating the approach using experimental measurements.